

CLOSURE PROPERTIES OF REGULAR LANGUAGES AND CONTEXT-FREE LANGUAGES (CFLs)

This table is one of the most important to memorize for Theory of Computation exams.

Operation	Regular Languages	Context-Free Languages
Union ($\cup$)	✓ Closed	✓ Closed
Concatenation ($\cdot$)	✓ Closed	✓ Closed
Kleene Star ($*$)	✓ Closed	✓ Closed
Intersection ($\cap$)	✓ Closed	✗ NOT Closed
Difference ($-$)	✓ Closed	✗ NOT Closed
Complement (Inverse)	✓ Closed	✗ NOT Closed
Reversal (L^R)	✓ Closed	✓ Closed
Inverse Homomorphism	✓ Closed	✓ Closed

1. REGULAR LANGUAGES — THE “SAFE ZONE”

Regular languages are extremely robust. Almost all standard operations performed on them result in another regular language.

- Union, Concatenation, Kleene Star

These are the core operations of regular expressions.

- Intersection

If M_1 and M_2 are DFAs for L_1 and L_2 , a product DFA can be constructed to accept $L_1 \cap L_2$.

- Complement

Take a DFA and flip final states to non-final and vice versa.

- Difference

$$L_1 - L_2 = L_1 \cap \bar{L}_2$$

Since regular languages are closed under intersection and complement, they are also closed under difference.

2. CONTEXT-FREE LANGUAGES — THE “DANGER ZONE”

Context-Free Languages are NOT closed under Boolean operations.

- Union, Concatenation, Kleene Star
Closed. Easy CFG constructions exist.

Example (Union):

$$S \rightarrow S_1 \mid S_2$$

- Intersection — NOT Closed

Classic counterexample:

$$L_1 = \{ a^n b^n c^m \} \text{ (matches a and b)}$$

$$L_2 = \{ a^m b^n c^n \} \text{ (matches b and c)}$$

$$L_1 \cap L_2 = \{ a^n b^n c^n \}$$

$\{ a^n b^n c^n \}$ is NOT a CFL.

Therefore, CFLs are NOT closed under intersection.

- Complement — NOT Closed

If CFLs were closed under complement, then by De Morgan's Laws:

$$L_1 \cap L_2 = (\bar{L}_1 \cup \bar{L}_2)^c$$

This would imply closure under intersection — which is false.

Interesting fact:

$\{ ww \mid w \in \{a, b\}^* \}$ is NOT a CFL,
but its complement IS a CFL.

- Difference — NOT Closed

$$L_1 - L_2 = L_1 \cap \bar{L}_2$$

Since complement is not closed, difference is also not closed.

SPECIAL CASE

$$\text{CFL} \cap \text{Regular} = \text{CFL}$$

Context-Free Languages ARE closed under intersection
with Regular Languages.